Ex 1: Data Preprocessing and Cleaning

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence
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Ex1: Data Preprocessing and Cleaning

Problem Statement:
A telecommunications company is experiencing customer churn (customers leaving the service). Before building a machine learning model to predict churn, the data science team must clean and preprocess the customer data to ensure high-quality input for analysis.

You are a Data Scientist working for the company. Your task is to preprocess the customer dataset by handling missing values, removing inconsistencies, transforming categorical features, scaling numerical attributes, and visualizing important relationships.

Learning Objective:
The objective of this experiment is to enable students to understand and perform the essential steps involved in data preprocessing. This experiment will provide a strong foundation for preparing high-quality data for subsequent machine learning and predictive analytics tasks.

Tasks to Perform:

1. Import the required Python libraries.
2. Load the Telco Customer Churn Dataset into a Pandas DataFrame. And display First 10 records, Last 10 records, Shape of the dataset, Column names.
3. Explore the Dataset and display Dataset information, Summary statistics, Number of missing values, Number of duplicate records
4. Clean the data
    a. Remove duplicate customer records.
    b. Replace missing values
        i. Numerical columns → Median
        ii. Categorical columns → Mode
    c. Verify that no missing values remain.
5. Encode categorical columns: Gender, Partner, Dependents, Contract, PaymentMethod, InternetService, Churn. Use appropriate encoding techniques.
6. Standardise the numerical features: Tenure, MonthlyCharges, TotalCharges, using StandardScaler.
7. Plot Distribution of MonthlyCharges, Distribution of Tenure.
8. Create a Count Plot showing Contract vs Churn.
9. Draw a Box Plot of MonthlyCharges grouped by Churn.
10. Display the Correlation Heatmap of all numerical features.

Aim

To clean and preprocess the Telco Customer Churn dataset by handling missing values, removing duplicates, encoding categorical features, scaling numerical attributes and visualising key relationships, so that the data is ready for churn prediction.

Procedure

1. Import pandas, numpy, matplotlib, seaborn and scikit-learn.
2. Load the Telco Customer Churn CSV into a DataFrame; display the first 10 and last 10 records, shape and column names.
3. Explore the data: info(), describe(), missing-value count and duplicate count.
4. Remove duplicate records; fill numerical missing values with the median and categorical missing values with the mode; verify no missing values remain.
5. Encode Gender, Partner, Dependents, Contract, PaymentMethod, InternetService and Churn using Label Encoding / One-Hot Encoding.
6. Standardise Tenure, MonthlyCharges and TotalCharges using StandardScaler.
7. Plot distributions of MonthlyCharges and Tenure, a count plot of Contract vs Churn, a box plot of MonthlyCharges by Churn, and a correlation heatmap.

Code

```
#Mohammed Fayyad(241801165)
import pandas as pd, numpy as np
import matplotlib.pyplot as plt, seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 2. Load data
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
print(df.head(10)); print(df.tail(10))
print(df.shape); print(df.columns.tolist())

# 3. Explore
df.info()
print(df.describe())
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

# 4. Clean
df = df.drop_duplicates()
num_cols = df.select_dtypes(include=np.number).columns
cat_cols = df.select_dtypes(include="object").columns
for c in num_cols:
    df[c] = df[c].fillna(df[c].median())
for c in cat_cols:
    df[c] = df[c].fillna(df[c].mode()[0])
```

```
print("Missing after cleaning:", df.isnull().sum().sum())

# 5. Encode
raw = df.copy()                        # keep un-encoded copy for plots
le = LabelEncoder()
for c in ["gender", "Partner", "Dependents", "Churn"]:
    df[c] = le.fit_transform(df[c])
df = pd.get_dummies(df, columns=["Contract", "PaymentMethod", "InternetService"],
drop_first=True)

# 6. Scale
scaler = StandardScaler()
cols = ["tenure", "MonthlyCharges", "TotalCharges"]
df[cols] = scaler.fit_transform(df[cols])

# 7-10. Visualisation
sns.histplot(raw["MonthlyCharges"], kde=True); plt.title("Monthly Charges");
plt.show()
sns.histplot(raw["tenure"], kde=True); plt.title("Tenure"); plt.show()
sns.countplot(x="Contract", hue="Churn", data=raw); plt.show()
sns.boxplot(x="Churn", y="MonthlyCharges", data=raw); plt.show()
sns.heatmap(df.select_dtypes(include=np.number).corr(), cmap="coolwarm"); plt.show()
```

Result

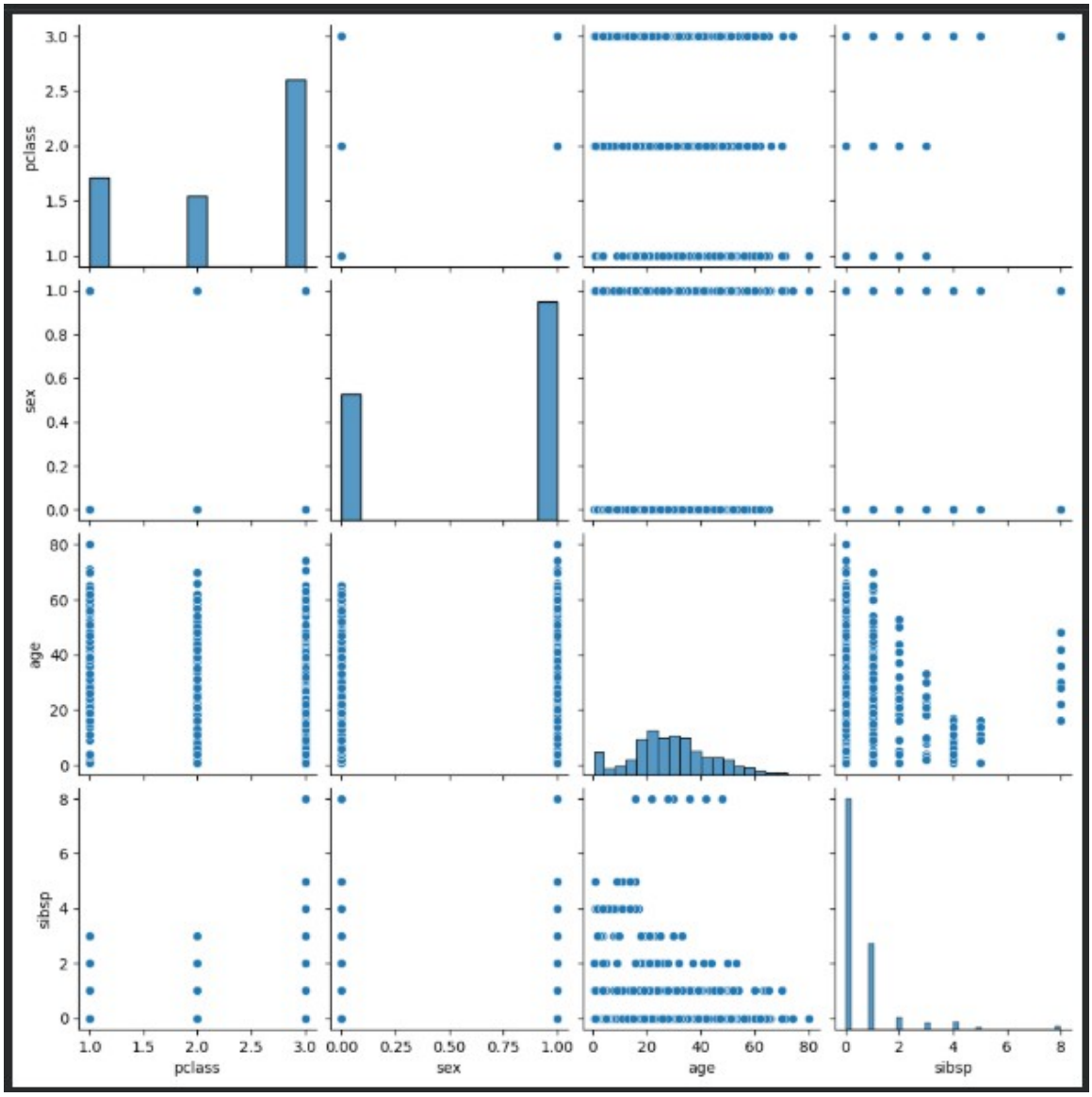
The dataset was cleaned (duplicates removed, no missing values remaining), categorical features were encoded, numerical features were standardised, and the required plots were produced. The preprocessed data is ready for modelling.

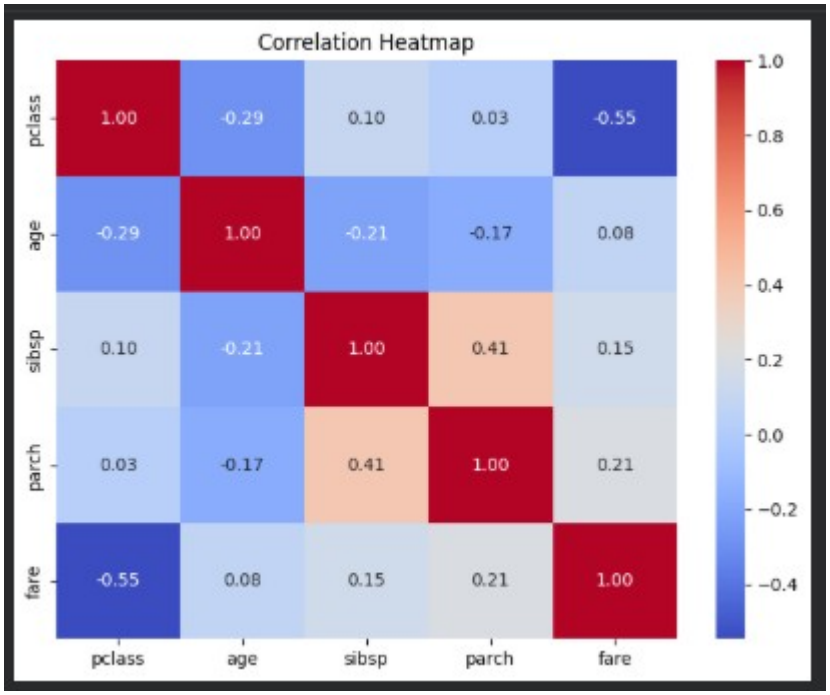
```
First 5 Rows:
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None

Missing Values:
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
```


pclass
sex
age
sibsp
pclass
sex
age
sibsp


Correlation Heatmap
pclass
age
sibsp
parch
fare
1.00
-0.29
0.10
0.03
-0.55
-0.29
1.00
-0.21
-0.17
0.08
0.10
-0.21
1.00
0.41
0.15
0.03
-0.17
0.41
1.00
0.21
-0.55
0.08
0.15
0.21
1.00
pclass
age
sibsp
parch
fare
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4